



RAINEY RACING

Kart Setup & Race-Day Book

Dirt-Oval Clone & Predator Box-Stock Racing

Home Track: Camden Speedway Kart Racing • Camden, TN

#10X	Predator 212 Box-Stock	Travis — Driver
#3	Clone Class	Ronnie — Driver
Crew	Engine & Chassis	Ronnie Sr. — Crew Chief

“Whatever you do, work at it with all your heart, as working for the Lord.” — Colossians 3:23

Team Reference Binder — Keep This Book in the Trailer

Edition: July 2026 • Driver / Kart: _____ • Book #: _____

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1. Team, Track & How to Use This Book

Who's who

Kart / Role	Class	Name	Notes
#10X	Predator 212 Box-Stock + Run What You Brung	Travis	Brand-new box-stock motor (June 2026)
#3	Clone Class + Run What You Brung	Ronnie	35 years of dirt racing
Crew Chief	Engine & chassis setup	Ronnie Sr.	Keeps both karts race-ready

Home track

- **Track:** Camden Speedway Kart Racing
- **Address:** 500 US-70 Business, Camden, TN
- **Race day:** Sunday afternoons (the day after Camden's Saturday races). Dates follow Camden's schedule and can change — always confirm the week of.
- **Surface:** Clay dirt oval, turns left. It usually starts tacky in the morning and slicks off as the day goes on — your setup will need to change with it (see Section 9).

Our three classes in plain English

- **Predator 212 Box-Stock (#10X):** A single-cylinder 212cc Predator engine, kept close to stock (“box stock” = only the changes the rulebook allows). Affordable, reliable, and hard to beat when it's maintained right.
- **Clone Class (#3):** A 6.5-hp Honda-GX200-style “clone” engine class. Very close cousin to the Predator — nearly all the maintenance in this book applies to both.
- **RWYB:** Open / run-what-you-brung. Fewer restrictions — both karts run this too. Same maintenance; setup is up to you.

✓ Golden rule of this book

Change ONE thing at a time, write it down, then feel the difference on track.

A setup you can't repeat is a setup you don't own. The blank log sheets at the back are how you build a winning notebook over the season.

How to use this book

- Green/blue boxes are tips and “do-this” reminders. Gold and red boxes are warnings — don't skip them.
- Anywhere you see a blank line (_____), that's a number for YOUR kart. Fill it in the first time you find a setting you like, in pencil, so you can update it.
- The checklists (Sections 10–13) are meant to be photocopied or reprinted so you can physically check boxes each race day.

Team contacts & account numbers

Fill these in once and you'll never dig for them at the track.

Who / What	Name	Phone	Account # / Notes
Track office / race director			
Engine builder			
Chassis / kart dealer			
Tire supplier			
Parts (online)			
Parts (local)			
Tow / hauler help			
Nearest ER / urgent care			
Insurance			
Other			

2. Know Your Kart

A dirt-oval kart is simple on purpose. Here are the parts you'll talk about all season, grouped the way you'll actually work on them.

The main parts

Area	Parts	What it does
Frame / chassis	Main rails, seat, seat struts, nerf bars, front & rear bumpers	The steel skeleton. Flex in the frame is part of your setup.
Front end	Spindles, king pins, tie rods, steering shaft, wheel	Steers the kart. Holds your caster, camber and toe settings.
Rear end	Live rear axle, axle bearings, hubs, rear wheels	One solid axle (no differential) — both rear tires always spin together. This is why stagger matters.
Tires & wheels	Left/right fronts & rears, valve stems, beadlocks (if used)	Your #1 grip adjustment. Size difference = stagger.
Engine	Predator 212 or GX200 clone, carburetor, air filter, exhaust, fuel tank	The power. Air-cooled single-cylinder 4-stroke.
Drivetrain	Clutch, chain, driver (clutch) gear, axle sprocket, guard	Sends power to the axle. Gearing lives here.
Brakes	Rear disc, caliper, master cylinder, brake line, pedal	Usually rear-only hydraulic disc. Bleed it and check pads.
Controls	Throttle pedal & cable, brake pedal, steering wheel, kill switch	Where the driver lives. Kill switch must always work.

Your two engines at a glance

The Predator 212 and the GX200 clone are built around the same design. Treat them almost identically. Always confirm the exact numbers against your engine builder and the current Camden/class rulebook — the boxes below are typical starting points, not gospel.

Item	Predator 212 (#10X)	GX200 Clone (#3)
Type	Air-cooled OHV single, 4-stroke	Air-cooled OHV single, 4-stroke
Displacement	212 cc	196–200 cc
Oil capacity (approx.)	~18–20 oz — fill to dipstick line	~16–20 oz — fill to dipstick line
Spark plug	NGK (e.g., BPR6ES/BP6ES) — gap ~0.028–0.030"	Same family — gap ~0.028–0.030"
Valve lash (cold)	~0.004" intake & exhaust*	~0.004" intake & exhaust*
Fuel	Fresh pump gas (per rules)	Fresh pump gas (per rules)
Chain	#35 (typical)	#35 (typical)

**Valve lash varies by builder and rules — confirm your exact spec and write it here: Intake _____" Exhaust _____"*

Racer's dictionary (know these words)

Term	What it means
Tight / Push	The kart won't turn — the front end slides toward the wall. Not enough front grip (or too much rear).
Loose / Free	The back end wants to come around and spin — too much rotation. Not enough rear grip.
Bite / Hook up	How well the tires grab. "It won't hook up" = no forward drive off the corner.
Stagger	The right-rear tire being bigger around than the left-rear. Helps the kart turn left.
Roll-out	The distance a tire rolls in one full turn (how you measure stagger).
Crossweight / Wedge	The percentage of weight on the right-front + left-rear diagonal. Tunes tight vs. loose.
Left-side %	How much of the total weight sits on the two left tires. More helps left turns.
Tacky / Heavy	A moist, high-grip track (usually early). The kart hooks up hard.
Slick / Dry-slick	A dried-out, low-grip track (usually by feature time). You hunt for bite.
Valve lash	The tiny gap in the valve train, set with a feeler gauge, engine cold.
Stall / Engagement	The RPM where the clutch grabs and the kart starts to move.
Green / Feature	Green flag = go. The feature is the main, longer race that pays.

3. Parts List & Spares Inventory

This is what lives in the trailer so a broken part never ends your day. The “Have / Need” column is for a quick monthly stock-check — tick what you have, circle what to reorder.

⚠ Never leave the shop without these

Spare chain + several master links + chain breaker/tool (you've broken a chain mid-day before).

Spare motor-mount / motor-plate bolts + lock washers + thread-locker (a loose motor plate has cost you laps).

A full set of spare tires/wheels, and the tools to change gearing.

Kill switch spare or wire to jump it, and a spare spark plug already gapped.

Engine consumables

Item	Suggested qty in trailer	Have / Need
Engine oil (your race spec)	2–3 quarts	
Oil drain pan + funnel + fill bottle	1 each	
Air filter (spare) + filter oil	1–2 filters	
Pre-filter / outerwear cover	2	
Spark plugs (pre-gapped)	2–3	
Carburetor gasket / intake gasket set	1 set each engine	
Fuel filter(s)	2–3	
Head gasket	1 per engine	
Recoil / pull-starter rope	1	

Drivetrain spares

Item	Suggested qty	Have / Need
#35 chain (spare loops)	2	
Master links (#35)	6–10	
Clutch (spare complete)	1	
Clutch shoes / springs / drum	1 set	
Driver (clutch) gears — range of sizes	3–4 sizes	
Axle sprockets — range of sizes	3–5 sizes	
Axle key(s) & sprocket hardware	several	
Chain lube / wax (dirt-type, dry)	1–2 cans	

Chassis & steering spares

Item	Suggested qty	Have / Need
Tie rod + tie rod ends	1–2	
Spindle (spare)	1	
Axle bearings + cassettes/collars	2	
Seat bolts, nerf-bar & bumper hardware	assortment	
Steering shaft bearing / bushings	1	
Brake pads + pad hardware	1–2 sets	
Brake fluid (sealed) + bleed kit	1	
Throttle & brake cables/springs	1 each	
Kill switch + wire + connectors	1	

Tires, fluids & hardware

Item	Suggested qty	Have / Need
Race tires (matched to track/rules)	2+ full sets	
Mounted spare wheels	4	
Valve stems, cores, caps	assortment	
Lead ballast + mounting bolts	as needed for min. weight	
Nuts/bolts/washers organizer (SAE)	1 bin	
Safety wire + zip ties + cotter pins	assortment	
Thread-locker (blue) + anti-seize	1 each	
Duct/racer tape, RTV, WD-40, brake clean	1 each	
Rags, nitrile gloves, hand cleaner	plenty	

4. Tools & Trackside Kit

If a tool lives in the trailer, you'll have it when a bolt backs out on grid. Build this once, then just restock.

Core hand tools

- Socket set (SAE + metric), 1/4" & 3/8" drive
- Combination wrenches (SAE + metric)
- Ratchets + extensions + swivel
- Screwdrivers (flat + Phillips)
- Allen / hex key set (SAE + metric)
- Pliers: needle-nose, slip-joint, side-cutters
- Adjustable wrench + vise grips
- Ball-peen hammer + dead-blow / rubber mallet
- Utility knife + scissors + files (flat & round)

Kart-specific tools

- Chain breaker / chain tool + chain alignment tool
- Clutch removal tool + gear puller
- Torque wrench (in-lb and ft-lb) — for head bolts, axle, flywheel
- Feeler gauges (for valve lash) + spark-plug gap tool
- Tire pressure gauge (low-range, 0–20 psi, accurate)
- Stagger tape / roll-out tape (pi tape) or a tape measure
- Bead breaker + tire irons + bead sealer + hand pump/compressor
- Caster/camber gauge + toe plates (or string + tape)
- Kart stand / jack
- Spark tester (inline) + multimeter

Setup & measuring

- Kart scales (4-corner) — the fastest way to fix tight/loose
- Tire durometer (rubber hardness) + tire pressure gauge
- Pyrometer (tire temps) — reads what the tire is really doing
- Tape measure + machinist's square + notepad/pencil

Power, shop & comfort

- Cordless impact + drill + charged batteries
- Small generator or power station + extension cords + work light
- Air compressor (or CO2/hand pump) for tires
- EZ-up canopy, folding table, chairs, floor mat
- Fire extinguisher (ABC) — mounted and in reach
- Trash bags, paper towels, cooler, sunscreen, bug spray

→ Pre-load a "grid bag."

A small bag you carry to the staging line: tire gauge, 1/2" & spark-plug wrench, tape, zip ties, safety wire, rag, spare plug, and a marker. Fixes 90% of last-second problems without a trailer run.

5. Maintenance Schedule

Dirt is hard on karts. A kart that's maintained on a schedule is faster, safer, and cheaper than one that's fixed only when it breaks. Full how-to for each engine job is in Section 6; drivetrain in Section 7.

After EVERY race day

- ☐ Pressure-wash / blow off the whole kart while the mud is fresh (protect the carb intake and electrics).
- ☐ Change or check the oil (dirt-class engines like fresh oil often — many run it every race).
- ☐ Remove, clean and re-oil the air filter; check the intake boot and clamps for cracks/leaks.
- ☐ Check chain: wear, tension, and that the master link is solid. Re-lube.
- ☐ Check ALL motor-mount / motor-plate bolts and re-torque; look for anything backing out.
- ☐ Look over tires for chunks/cuts; note pressures; check wheels and lug/bead hardware.
- ☐ Wipe the chassis and look for cracks, bent nerf bars, loose seat bolts.
- ☐ Bleed/verify brakes feel firm; check pads.
- ☐ Top off fuel or drain per your storage habit; note anything to fix before next week.

Weekly / between races (in the shop)

- ☐ Re-torque head bolts and check valve lash (cold) — see Section 6.
- ☐ Inspect clutch: shoes, drum, engagement; grease per maker's instructions.
- ☐ Check axle bearings for play/roughness; check hubs and keys.
- ☐ Check steering: tie-rod ends, king pins, no slop in the wheel.
- ☐ Confirm caster/camber/toe haven't moved; check ride height.
- ☐ Go over every fastener with the torque list; safety-wire what should be wired.
- ☐ Restock the trailer from the spares list (Section 3).

Monthly / every few races

- ☐ Replace the air filter if it's been run hard on dry-slick (dust).
- ☐ Fresh spark plug; inspect the old one and read it (Section 6).
- ☐ Drain and clean the carburetor bowl; check the fuel filter.
- ☐ Repack / replace axle bearings if gritty; inspect the axle for straightness.
- ☐ Rotate/measure tires; log stagger and wear patterns.
- ☐ Full fastener audit + brake system check + fluid change if spongy.

Seasonal / off-season

- ☐ Send the engine for freshen-up/inspection with your builder (or do a full teardown per rules).
- ☐ Replace chain and clutch if worn; new bearings all around.
- ☐ Check the frame on a flat table for tweak/cracks; repair or replace nerf bars & bumpers.
- ☐ Fresh brake fluid, new pads, inspect rotor and caliper.
- ☐ Deep-clean and re-organize the trailer; refresh the whole spares kit.
- ☐ Review the season's setup logs — build your “baseline that wins at Camden.”

Maintenance log

Photocopy this or use the full-page version in Appendix D.

Date	Kart	Work done	Parts used	By

6. Engine Care (Predator 212 & Clone)

Both engines are simple 4-strokes. Keep clean oil in them, keep dirt OUT of them, and keep the valves and plug in spec — do that and they'll run all season. Always follow your class rules and engine builder for exact specs.

Oil — the #1 thing

- Use the oil your class/builder specifies. Change it OFTEN — many dirt racers change every race day because heat and dust break oil down fast.
- Warm the engine, drain from the plug, refill to the dipstick line. Do NOT overfill — it robs power and can blow seals.
- Capacity is roughly 18–20 oz (Predator) / 16–20 oz (clone) — but always trust the dipstick line, not the bottle.
- Catch old oil in a pan and recycle it. Note the change on your log.

My oil: _____ Change every: _____ race day(s)

Air filter — dirt's favorite way to kill an engine

⚠ On a dry-slick day, dust is the enemy

The tiniest air leak past the filter sands the cylinder and you lose power (or the engine). Every race day: clean/re-oil the filter, and make sure the filter adapter, clamp and intake boot are TIGHT and crack-free.

- Foam filter: wash in soapy water, dry fully, re-oil with foam-filter oil, squeeze out excess.
- Paper filter: replace when dirty — don't try to wash it.
- Run a pre-filter/outerwear sock on dirt; it buys you a lot of filter life.

Spark plug

- Gap to spec (about 0.028–0.030" / 0.7 mm unless your builder says otherwise).
- Carry a spare that's already gapped. Change plugs a few times a season.
- Read the old plug: light tan/gray = good; black & sooty = running rich; white/blistered = running lean or too hot.

Valve lash (cold)

Valve lash is the small clearance in the valvetrain. Too tight = burned valve and low power; too loose = noisy and down on power. Set it COLD, piston at top of the compression stroke (both valves closed).

1. Remove the valve cover. Rotate the engine by hand to top-dead-center on compression (both rockers loose).
2. Slide the feeler gauge between the rocker and valve stem. It should have a light drag.
3. Adjust the rocker nut until the gauge just drags, then lock it and recheck.
4. Do intake and exhaust, put the cover back with a good gasket, and log the numbers.

My valve lash: Intake _____ " Exhaust _____ " (confirm with builder/rules)

Carburetor & fuel

- Keep the bowl clean — grit in the carb causes bogs and flooding. Drain and clean it monthly (more if it acts up).
- Run a fuel filter and use fresh fuel. Old/stale fuel is a common “it ran bad” cause.
- Jetting/mixture affects how it runs in heat and humidity: black smoke & load-up = too rich; surging/lean & hot = too lean. Small changes only, and re-read the plug.
- After a dusty day, check that no dirt got past the filter into the carb throat.

New-motor break-in

You put a fresh box-stock motor in the #10X in June 2026 — protect that investment.

- Follow your builder's break-in exactly (heat cycles, first oil change timing, RPM limits).
- Change the break-in oil early — it carries the first metal from the rings/bearings.
- Re-torque the head and re-check valve lash after the first couple of heat cycles.

Engine torque quick-reference

⚠ Confirm every number with your engine builder / manual

The values below are typical for these engines but vary by part and rulebook. Write YOUR confirmed numbers in the blanks and use a torque wrench — guessing here cracks parts or backs bolts out.

Fastener	Typical range*	YOUR spec
Cylinder head bolts	~18–24 ft-lb (in a cross pattern)	_____
Flywheel nut	~40–60 ft-lb	_____
Connecting-rod bolts	per builder (in-lb, critical)	_____
Spark plug	~15–20 ft-lb (snug + gasket crush)	_____
Muffler / header bolts	~8–12 ft-lb + lock/anti-seize	_____
Motor mount / plate bolts	tight + blue thread-locker	_____

**Starting points only — your builder's numbers win.*

7. Drivetrain — Chain, Clutch & Gearing

This is where power gets to the ground — and where a small miss ends your day. You've lost laps to a broken chain and a loose motor plate before, so this section gets its own pre-race attention every single week.

Chain

- Type: #35 chain is typical for these karts. Keep a spare loop and plenty of master links in the trailer.
- Tension: a little slack, not tight. Aim for roughly 1/4"–1/2" of up-and-down play at the middle of the run. Too tight snaps chains and wears sprockets; too loose jumps off.
- Alignment: the clutch (driver) gear and axle sprocket must line up perfectly. A misaligned chain throws itself off — check with a straightedge.
- Master link: install the clip so the closed end faces the direction of travel. A backwards clip works loose and the chain comes apart.
- Lube: use a dry/wax chain lube for dirt — wet oily lube grabs grit and grinds the chain. Re-lube every race day.
- Replace when: it's stretched, kinked, or has tight spots. A cheap chain is cheaper than a DNF.

⚠ Your chain checklist before EVERY session

1) Tension correct (1/4"–1/2" play). 2) Sprockets aligned. 3) Master-link clip on and facing forward. 4) Lubed.

It takes 60 seconds and it's the difference between a finish and a walk back to the pits.

Clutch

- Centrifugal clutch: it grabs (engages) at a set RPM as the engine revs up. Keep the drum and shoes clean; grease only where the maker says (over-greasing makes it slip).
- Engagement/stall RPM: too low and it drags/overheats at idle; too high and it bogs off the corner. Tune to your track and gearing.
- Watch for: blue/burnt drum, glazed shoes, or a clutch that slips at speed — time to service or replace.

My clutch: _____ Engagement RPM: _____

Gearing — how you tune drive vs. top speed

Gear ratio = axle-sprocket teeth ÷ driver-gear teeth. Change one tooth and you change the whole feel off the corner.

If you want...	Change	Effect
More acceleration / drive off the corner (tight, tacky, or short track)	MORE teeth on the axle sprocket (or fewer on the driver)	Lower gear — quicker off the corner, less top end. Higher RPM at the flag stand.
More top speed (big, slick, fast track)	FEWER teeth on the axle sprocket (or more on the driver)	Higher gear — more top end, softer off the corner. Lower RPM at the flag stand.

- Tune by lap time AND engine RPM, not by feel alone. Note the RPM you pull down the straight and log it with the gear.

- Start from your baseline gear, then go one tooth at a time. Write down what the track was doing.

My baseline gear @ Camden: Driver _____ T Axle _____ T RPM _____

Motor mount / motor plate

The loose-plate lesson

A motor plate that walks loose pops the chain and can wreck a race (you've lived it). Every race day: check every mount/plate bolt with a wrench, use blue thread-locker, and consider safety-wiring or paint-marking the bolts so a glance tells you if one moved.

- Paint-mark trick: after torquing, draw a paint pen line across each bolt head onto the plate. If the line breaks, the bolt moved — you'll spot it instantly.

8. Chassis & Dirt Setup Basics

→ The one idea behind all dirt-oval setup

Your kart only turns LEFT, and it has ONE solid rear axle (no differential — both rear tires always turn together). Every setting below exists to help the kart rotate left and put power down on a track that keeps changing all day.

Stagger (your biggest turning tool)

- The right-rear tire is bigger around than the left-rear. That built-in size difference helps the solid axle roll around a left turn.
- Measure roll-out (distance per turn) or circumference with a stagger tape. RR minus LR = your stagger.
- MORE stagger = turns left easier (frees the kart / helps a push). TOO much = loose and darty.
- Typical small-track starting stagger is roughly 1/2"–2" — but your tires and track set the real number. Log it.

My stagger: Rear RR _____ LR _____ = _____ | Front RF _____ LF _____ = _____

Tire pressure

- Lower pressure = bigger contact patch = more grip (great for finding bite on slick) — until it gets too low and rolls/gets greasy.
- Higher pressure = less grip, can free the kart up or help on a heavy/tacky track.
- Dirt kart pressures run LOW and depend heavily on your tire brand and prep — start where your tire maker/track buddies suggest and log every change. Set them cold; note hot pressures too.

My cold pressures: LF _____ RF _____ LR _____ RR _____ psi

Weight, left-side & crossweight

- Hit the class minimum weight WITH the driver in the seat. Add lead low and to the LEFT to help left turns — never above the seat, never loose.
- Left-side %: more weight on the two left tires helps the kart turn. Crossweight/wedge (RF + LR diagonal) tunes tight vs. loose (Section 9).
- Scales are the fastest way to see this. If you don't have them yet, they're the best setup tool you can buy.

Seat position (a primary dirt tuning tool)

- Where the driver's weight sits changes rear grip a LOT on a kart. Moving the seat back/down generally adds rear bite; forward/up frees it.
- Move in small steps and only one direction at a time — then log it. It's a big lever, so a little goes a long way.

Track width (front & rear)

Effects vary by chassis, so test and log — but the common dirt-oval tendencies are:

- Wider REAR track = frees the kart up (less side bite). Narrower REAR = more rear bite.
- Wider FRONT = more steering response/front bite. Narrower FRONT = calmer steering.

Front end: caster, camber, toe

Setting	What it does	Starting idea
Caster	More caster adds steering weight and helps the kart bite and turn in.	Set per chassis manual; note baseline.
Camber	Tilts the tire for even contact/temps in the corner.	A little negative is common; check tire temps.
Toe	Toe-out helps the kart turn in; too much scrubs speed.	Slight toe-out (~1/16"–1/8" total) is a common start.

Axle stiffness & hub length (rear grip)

- Softer axle OR longer rear hubs = MORE rear grip (good when the track goes slick).
- Stiffer axle OR shorter hubs = LESS rear grip / frees the kart (good when it's tacky and hooked up too hard).
- If your class allows different axles/hubs, they're a great way to chase a slicking track without touching stagger.

Write down your baseline

Once you find a setup that works at Camden, record it here and on Appendix A. That baseline is your starting point every week.

Setting	Baseline	Setting	Baseline
Rear stagger		Left-side %	
Front stagger		Crossweight %	
LF / RF pressure		Seat position	
LR / RR pressure		Rear width	
Caster		Front width	
Camber		Axle / hubs	
Toe		Gear (driver/axle)	

9. Dirt-Track Tuning Guide

This is the heart of the book: how to make the kart do what you want as the track changes through the day. Every kart is a little different, so treat these as directions to test — change ONE thing, log it, feel it.

Reading the track through the day

Time of day	Track condition	What the kart wants
Morning / hot laps	Tacky, heavy, high grip	It may push (tight) because grip is high — you can free it up and gear for drive.
Midday / heats	Drying, changing lap to lap	Small tweaks between sessions; watch where the grip is on the track.
Afternoon / feature	Slick, dry-slick, low grip	Now you HUNT for bite: find grip and keep it hooked up without going too tight.

The two problems: TIGHT vs. LOOSE

TIGHT (push): the front won't turn and slides toward the wall. You need the kart to rotate more.

LOOSE (free): the rear steps out and wants to spin. You need more rear grip / less rotation.

Fix-it table — kart is TIGHT / pushing

Lever	Do this	Why
Stagger	ADD rear stagger	Helps the axle roll around the left turn.
Crossweight	TAKE OUT crossweight (less wedge)	Frees the front so it turns.
Seat / weight	Move a little weight forward / free the rear	Less rear grip = more rotation.
Rear grip	Stiffer axle or shorter hubs	Frees the rear so the kart rotates.
Front tires	A touch more front grip (adjust front pressure)	More steering bite up front.
Rear width	Widen the rear track (test)	Common tendency: frees the kart.

Fix-it table — kart is LOOSE / free

Lever	Do this	Why
Stagger	REDUCE rear stagger	Less roll = calmer rear.
Crossweight	ADD crossweight (more wedge)	Tightens the kart up.
Seat / weight	Move a little weight rearward	Adds rear grip.
Rear grip	Softer axle or longer hubs	More rear bite.
Tire pressure	Lower rear pressure for more grip	Bigger contact patch at the rear.
Rear width	Narrow the rear track (test)	Common tendency: adds side bite.

Won't hook up / no forward drive (slick track)

- Lower tire pressures to grow the contact patch (front and/or rear).
- Add rear grip: softer axle / longer hubs / a bit more rear weight or seat back.
- Re-gear so you're not spinning the tires off the corner (don't over-gear the bottom).
- Check tire prep/compound is right for a slick track and that tires aren't worn out.

Tacky-track strategy (early)

- Grip is high, so the kart can get tight — free it up (more stagger, less wedge, stiffer axle).
- You can usually gear for more drive; watch that you don't overheat the clutch or spin.

Slick-track strategy (feature)

- Grip is low, so find bite everywhere — lower pressures, softer axle/longer hubs, seat/weight for rear grip.
- Smoothness wins: the driver who keeps the tires hooked up beats the one who overdrives and spins.

✓ The rule that makes you fast

ONE change at a time. Write down the track condition, the change, and what it did. In a few race days you'll have a Camden playbook nobody can copy — that's how the fast teams do it.

Driver feedback words (so Travis & Ronnie Sr. speak the same language)

Driver says...	Crew hears...
"It won't turn in the middle"	Tight / push — free it up (tight table).
"The rear steps out on entry/exit"	Loose — add rear grip (loose table).
"No drive off the corner"	Won't hook up — pressures/axle/gear.
"Inside rear lifts / it bicycles"	Too much crossweight or grip imbalance — check wedge & stagger.
"Good in the middle, tight off"	Rear grip/gear off the corner — small change only.
"Fast but on edge / darty"	Maybe too much stagger or too free — calm it a touch.

10. Pre-Race-Day Checklist (at home / night before)

Do this the night before so race morning is calm. Photocopy this page for each kart.

Engine

- ☐ Oil changed / level correct
- ☐ Air filter clean, oiled, and sealed tight
- ☐ Fresh/gapped spark plug in; spare in the bag
- ☐ Valve lash checked (if due)
- ☐ Carb clean, fuel filter good, fresh fuel ready
- ☐ Head bolts & muffler bolts re-torqued

Drivetrain

- ☐ Chain: tension, alignment, master-link clip forward, lubed
- ☐ Spare chain + master links packed
- ☐ Clutch clean & checked
- ☐ Correct baseline gear on; spare sprockets/driver packed
- ☐ Motor-mount / plate bolts torqued + paint-marked

Chassis, tires & brakes

- ☐ Baseline setup on the kart (stagger, pressures, width, caster/camber/toe)
- ☐ Tires: correct set mounted, no chunks, pressures set, spares loaded
- ☐ Ballast bolted low & left; kart makes minimum weight with driver
- ☐ Steering tight, nothing loose; seat bolts tight
- ☐ Brakes firm, pads good, fluid topped
- ☐ Every fastener checked; safety wire where needed

Driver gear

- ☐ Helmet (in-date) + fresh shield/tear-offs
- ☐ Neck brace, suit, gloves, shoes, rib protector
- ☐ Backup base layers / rain gear
- ☐ Water + snacks + electrolytes

Trailer & paperwork

- ☐ Spares kit restocked (Section 3)

- ☐ Tools & grid bag packed (Section 4)
- ☐ Fuel + oil + chemicals loaded
- ☐ Scales / pyrometer / gauges packed
- ☐ Membership / license / transponder
- ☐ Cash for pit passes & entry; phone charged
- ☐ Canopy, chairs, cooler, sunscreen, first-aid kit
- ☐ Fire extinguisher on board & charged

11. At-the-Track Checklist (race-day flow)

Follow the day in order. Check boxes as you go so nothing gets missed in the rush.

1) Arrive & unload

- ☐ Park & set up: canopy, tools, scales, fire extinguisher in reach
- ☐ Unload karts; quick walk-around for anything shaken loose in the trailer
- ☐ Find the sign-in, tech line, and driver's meeting time

2) Sign-in & tech

- ☐ Sign in / pay pit passes & entry; get transponder if used
- ☐ Pass tech inspection (weight, safety, class rules)
- ☐ Attend the drivers' meeting — note flag/procedure/track changes

3) Before hot laps / practice

- ☐ Fuel in, oil level good, kill switch works
- ☐ Chain tension + master link + alignment (60-second check)
- ☐ Tire pressures set cold; baseline setup confirmed
- ☐ Bolts spot-checked; nothing leaking
- ☐ Driver geared up; belts/collar snug; seat comfortable

4) Practice — what to test

- ☐ Is it tight or loose? Where on the track? (Sections 9)
- ☐ Note gear/RPM down the straight
- ☐ Check hot tire pressures & temps right after coming in
- ☐ Make ONE change, log it, go back out

5) Between sessions (heats → feature)

- ☐ Re-check chain, motor-plate bolts, tire pressures every time out
- ☐ Adjust for how the track is changing (tacky → slick)
- ☐ Re-fuel; wipe the kart; quick bolt check
- ☐ Log the setup you're running for the heat/feature

6) Grid / pre-race (heat AND feature)

- ☐ Fuel enough for the race length

- ☐ Chain + master link + motor-plate bolts — last look
- ☐ Tire pressures correct
- ☐ Kill switch works; nothing dragging
- ☐ Driver: belts, collar, gloves, shield clear, hydrated

⚠ Same three things, every time out

Chain (tension + clip forward) · Motor-plate bolts (tight + paint mark intact) · Tire pressures. These are the three that have cost you races before — make them a reflex.

12. Post-Race & Load-Out

Right after the feature

- ☐ Let the engine cool; don't blast a hot engine with cold water
- ☐ Quick look for damage, leaks, bent parts, chunked tires
- ☐ Grab your final hot pressures/temps and jot the day's notes while it's fresh
- ☐ Collect trophies/payout; return transponder

Load-out

- ☐ Strap karts down (front & rear); chock/secure everything that moves
- ☐ Pack tools & spares; count them so nothing's left in the dirt
- ☐ Secure fuel & fluids upright; close the canopy
- ☐ Walk your spot — leave it cleaner than you found it
- ☐ Fill in the Setup Log (Appendix A) before you forget the details

Team debrief (2 minutes)

- What was the kart doing at the end? What change helped most? What do we try next week?
- What broke or worried you? Add it to the shop list and the spares reorder.

13. Race-Day Packing List (trailer load-out)

The master “did we bring it?” list. Print it and check it as you load the trailer.

Karts & spares

- ☐ Both karts
- ☐ Spare tires/wheels (multiple sets)
- ☐ Spares kit (Section 3)
- ☐ Ballast + bolts
- ☐ Fuel + oil + chain lube + chemicals

Tools & setup

- ☐ Toolbox + kart-specific tools
- ☐ Torque wrench + feeler & gap gauges
- ☐ Scales, pyrometer, durometer
- ☐ Tire gauge + air source
- ☐ Grid bag
- ☐ Work light + power/generator

Driver & safety

- ☐ Helmet + neck brace + suit + gloves + shoes + rib protector
- ☐ Backup layers / rain gear
- ☐ First-aid kit
- ☐ Fire extinguisher
- ☐ Sunscreen, bug spray, hats

People & paperwork

- ☐ License / membership / transponder
- ☐ Cash for pits & entry
- ☐ This book + pens + setup logs
- ☐ Phone + charger / power bank
- ☐ Canopy, chairs, table, cooler
- ☐ Water, snacks, electrolytes

14. Safety Gear & First Aid

Nothing in this book matters more than the driver coming home safe. Check gear rules with Camden/your class — requirements can change, and a helmet or suit that isn't in-date can keep you from racing.

Driver gear

Gear	What to check
Helmet	Correct, in-date safety rating (e.g., Snell); no cracks; shield/tear-offs clean; fits snug.
Neck brace / collar	Correct type for karting; not cracked; worn every time on track.
Suit / jacket	Abrasion-rated per rules; zippers & straps good; fits over layers.
Gloves	Full-finger, good grip, no holes.
Shoes	Closed, over-the-ankle, thin sole for pedal feel.
Rib protector	Recommended for kart racing — saves a lot of bruised/cracked ribs.

- Replace a helmet after any real impact, and don't buy used helmets — you can't see their history.
- Store gear out of the sun and heat; UV and heat kill helmets and suits.

Fire & pit safety

- ☐ Keep a charged ABC fire extinguisher within reach at the kart AND in the trailer.
- ☐ No smoking/open flame near fuel; fuel on the ground, upright, capped.
- ☐ Kids and spectators clear of running karts and the hot pit lane.
- ☐ Know where the track's medical/EMT and the exit are.

First-aid kit contents

Category	Items
Wounds	Bandages, gauze, medical tape, antiseptic wipes, antibiotic ointment, burn gel
Tools	Trauma shears, tweezers, gloves, instant cold pack, elastic (ACE) wrap
Meds	Pain reliever, antihistamine, electrolyte tabs (per family needs)
Heat/sun	Sunscreen, aloe, extra water, shade — heat is the most common race-day problem
Info card	Emergency contacts, allergies, blood type, insurance — kept in the kit

Heat & hydration

- Start drinking water before you're thirsty; add electrolytes on hot Sundays.
- Cool the driver between sessions — shade, wet towel, sit down. Dizziness/cramps/nausea = stop and cool off now.

15. Troubleshooting

Fast field fixes for the problems you'll actually hit. Work top to bottom — the most common cause is listed first.

Engine won't start / hard to start

Check	Fix
Fuel	Fuel in tank? Valve on? Fresh fuel? Filter not clogged? Carb bowl has fuel?
Spark	Kill switch on/working? Plug wire on? Pull the plug — is there spark? Gap right?
Plug	Fouled/wet plug — swap in the pre-gapped spare.
Choke	Choke set correctly for a cold vs. warm engine.
Air	Filter not packed with mud; intake clear.
Compression	Slow, weak pull with no 'pop' = check valve lash / bigger engine issue.

Runs but bogs / won't rev / dies

Symptom	Likely cause & fix
Bogs off the corner	Clutch engagement too high, gear too tall, or lean — check clutch & jetting.
Loads up / black smoke / fouls plug	Too rich — lean the mixture a touch; check choke & air filter oil (over-oiled).
Surges / cuts out on the straight / hot & lean	Too lean or fuel starvation — check fuel flow, filter, vent, and mixture.
Sputters after a dusty run	Dirt in the carb — clean the bowl; check the filter seal.
Dies at idle	Idle/clutch drag or fuel — check idle set & clutch.

Overheating / down on power late

- Cooling fins & shroud clean (mud packs them and cooks the engine).
- Oil level correct and not overfilled; oil fresh.
- Not running too lean (lean = hot). Re-check the plug color.
- Clutch not slipping/dragging and heating up.

Chain problems

Problem	Fix
Chain came off	Check tension (1/4"–1/2"), sprocket alignment, and worn sprockets. Replace a stretched chain.
Chain broke	Install a fresh loop + new master link (clip forward). Inspect sprockets for damage.

Master link keeps failing	Clip installed backwards or reused — always fit a NEW link, closed end facing travel.
Motor plate/chain both loose	Re-torque plate bolts + thread-locker + paint-mark; reset chain tension after.

Clutch problems

- Slips at speed / blue drum = glazed or worn shoes — service or replace.
- Won't drive off idle = engagement too high or shoes/springs worn.
- Drags at idle / kart creeps = engagement too low or over-greased — clean & set.

Handling problems

Go to Section 9 and use the TIGHT or LOOSE fix-it table. Quick reminders:

- Won't turn (tight): add stagger, take out wedge, free the rear.
- Spins/loose: reduce stagger, add wedge, add rear grip.
- No forward bite: lower pressures, softer axle/longer hubs, check gear.

Vibration / strange noise — stop and check

- New vibration = loose motor mount, bent axle, wheel/hub, or clutch — DON'T keep running it.
- Grinding = bearing (axle or wheel). Squeal = brake or belt/clutch.
- If you can't find it fast, sit out the session. A cheap part now beats a big repair later.

Appendix A. Race-Day Setup Log (make copies)

Fill one out every race day. Over a season these pages become your winning playbook for Camden.

Date: _____ Track: _____ Class: _____ Kart #: _____

Weather / Temp: _____ Track condition (circle): TACKY / DRYING / SLICK

Tires

Corner	Brand / Compound	Size / Roll-out	Cold psi	Hot psi	Temp
LF					
RF					
LR					
RR					

Rear stagger (RR – LR): _____ Front stagger (RF – LF): _____

Chassis & weight

Setting	Value	Setting	Value
Total weight		Seat position	
Left-side %		Rear width	
Crossweight %		Front width	
Ballast / where		Caster / Camber / Toe	
Axle / hubs		Ride height	

Engine, gearing & clutch

Driver T	Axle T	Ratio	Straightaway RPM	Clutch / engagement

Results & notes

Heat: Start _____ Finish _____ Feature: Start _____ Finish _____

Handling (was it tight/loose, and WHERE on the track?):

Changes I made and what they did:

Next time try:

Appendix B. Gearing & RPM Log

Track how each gear worked so you can nail the setup faster next time.

Date	Track cond.	Driver T	Axle T	Ratio	RPM	How it drove / result

Appendix C. Maintenance Log

Every job, every part, every kart. This is your service history.

Date	Kart	Hours / races	Work done	Parts used	By

Appendix D. Season Results Tracker

Log every Sunday here — then it's easy to post the recap on raineymotorsports.com after the checkered flag.

Date	Kart / Class	Heat	Feature	Notes / highlights

Appendix E. Baseline & Torque Quick Card

Fill this in once you've dialed the karts. Snap a photo and keep it on your phone for the track.

Winning baseline @ Camden

Setting	#10X Predator	#3 Clone
Rear stagger		
Cold pressures (LF/RF/LR/RR)		
Left-side % / Cross %		
Seat position		
Rear / front width		
Caster / Camber / Toe		
Axle / hubs		
Gear (driver/axle)		
Clutch / engagement RPM		

Confirmed torque specs (from your builder)

Fastener	Torque	Fastener	Torque
Head bolts		Spark plug	
Flywheel nut		Muffler bolts	
Rod bolts		Motor-plate bolts	
Axle / hub		Clutch bolt	

Built for Rainey Racing — race hard, stay safe, and give God the glory. See you at Camden.